Self-Organized Criticality in a Two-Good Barter Economy: An Agent-Based Study of Avalanche Dynamics on Small-World Networks

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Abstract

We investigate whether a simple two-good barter economy, in which agents must continuously acquire two perishable resources to survive, exhibits the statistical signatures of Self-Organized Criticality (SOC). A population of $N = 100$ agents is placed on a Watts–Strogatz small-world network and engages in local pairwise trading of apples and oranges. Agents consume one unit of each good periodically; failure to consume results in death, followed by respawning after a fixed delay. Resources are injected into the system at a rate approximately equal to population demand, providing the slow drive required for SOC. We define avalanches as clusters of agent deaths occurring within one time-step of one another and measure their size distribution over pooled simulations. The avalanche-size distribution exhibits an approximately power-law regime over roughly one decade with exponent $\tau \approx 2.2$–2.8, and the exponent decreases monotonically with increasing network connectivity k. The variance of the avalanche-size distribution grows with k, providing a susceptibility-like signature of criticality. While the limited dynamic range precludes a definitive SOC claim, the results demonstrate that local trade networks with resource constraints can produce heavy-tailed failure statistics that depend systematically on network topology.

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1 Background and Literature Review

The study of Self-Organized Criticality (SOC) has evolved significantly since its introduction by Bak, Tang, and Wiesenfeld (1987). SOC systems are now recognized across domains including geophysics, biology, neuroscience, and economics. A key feature of such systems is the emergence of scale-invariant behaviour without external tuning.

In economics, several agent-based models have demonstrated that market crashes, wealth distributions, and trading dynamics can exhibit heavy-tailed behaviour. Models such as the Minority Game, percolation-based financial models, and network contagion models all highlight how local interactions can lead to global instability.

The work of Ribeiro et al. (2011) is particularly relevant, as it demonstrates SOC behaviour in economic systems with consumption constraints. Their model motivates the present study, which simplifies the framework while retaining core mechanisms such as resource constraints, local interactions, and stochastic driving.

In parallel, network science has shown that topology plays a crucial role in dynamical processes. Small-world networks, introduced by Watts and Strogatz, balance clustering and short path lengths, making them ideal for studying propagation phenomena such as cascades and avalanches.

This report builds upon these foundations to explore whether a minimal barter economy can exhibit SOC-like behaviour.

2 Introduction to Complexity Science

Complexity science is the interdisciplinary study of systems composed of many interacting components whose aggregate behaviour cannot be straightforwardly deduced from the behaviour of their individual parts. Such systems—ranging from colonies of ants and networks of neurons to financial markets and ecosystems—exhibit emergent properties that arise from the nonlinear coupling of simple local rules. Complexity science borrows tools from statistical physics, dynamical systems, network theory, and computer science to ask how order, structure, and characteristic patterns appear in systems that have no central controller and no globally specified goal.

A distinguishing feature of complex systems is that they frequently operate near the boundary between order and disorder. At this boundary, small perturbations can propagate through the system in highly nonlinear ways, producing responses whose magnitudes span many orders of magnitude. The statistics of such responses are often described by power-law distributions, which are scale-free: no particular event size is privileged, and events of all sizes contribute meaningfully to the dynamics. This absence of a characteristic scale is in stark contrast to the bell-shaped distributions familiar from independent random processes, where events far from the mean are exponentially rare.

2.1 Self-Organized Criticality

Self-Organized Criticality (SOC) is a concept introduced by Bak, Tang, and Wiesenfeld in 1987 to describe dynamical systems that spontaneously evolve toward a critical state without the need for fine-tuning of control parameters. The paradigmatic example is the Bak–Tang–Wiesenfeld sandpile: grains of sand are added one at a time to a pile, which progressively steepens until avalanches of all sizes become possible. The size distribution of these avalanches follows a power law, indicating that the pile has self-organized into a critical state at the edge of stability. SOC has since been invoked to explain phenomena as diverse as earthquake magnitudes (the Gutenberg–Richter law), forest fires, neuronal firing avalanches, and fluctuations in financial markets.

2.2 Key Terms

Noise. In the context of complexity science, noise refers to the small, random perturbations that drive a system away from any particular configuration. In SOC systems, noise plays the role of a slow external drive: a continuous trickle of input that keeps the system near its critical threshold. The noise is not the phenomenon of interest itself, but the stochastic input that reveals the system's response function. In the sandpile, noise is the location at which the next grain of sand is dropped. In our barter economy, noise is the random arrival of fruit drops into the population—random in timing, amount, location, and type.

Avalanche. An avalanche is a discrete cluster of cascading events, triggered by a small perturbation and terminating when the system returns to a (meta)stable configuration. The *size* of an avalanche is the number of elementary events it contains. In the sandpile, avalanche size is the number of toppled cells. In epidemiology, it is the number of infected individuals in an outbreak. In our model, we define an avalanche as a cluster of agent deaths occurring within one tick of one another: a triggering death whose propagation through the trade network causes subsequent failures. The statistical distribution of avalanche sizes is the principal diagnostic of criticality.

Connectivity. Connectivity describes how the components of a system are coupled to one another. In a networked system, it is captured by the mean degree k—the average number of neighbours each node possesses. Connectivity governs how far, and how fast, a perturbation can propagate. Sparse networks isolate events; dense networks couple every part of the system to every other. Critical behaviour, when it exists, typically emerges in an intermediate regime where stress can travel but does not immediately engulf the entire system.

3 System Description

Our system is a stylised barter economy consisting of N autonomous agents that exchange two perishable goods—apples and oranges—to sustain themselves. Each agent is endowed with private stocks of both fruits, measured in integer units, and must consume one apple and one orange periodically in order to survive. The agents do not communicate linguistically;

they interact only by offering and accepting trades with their immediate neighbours on a fixed network. The network topology is generated using the Watts–Strogatz small-world model with rewiring probability $p = 0.1$ and mean degree k, which is the principal control parameter we vary in the experiments.

The state of each agent comprises its apple count, its orange count, a flag indicating whether it is currently alive, and a timer specifying when it will next need to consume. At the population level, the system additionally tracks a list of death events and the identities of agents scheduled to respawn. The global dynamics proceed in discrete time steps ("ticks"), and at each tick the following operations are performed in order: respawning of eligible dead agents, slow drive via stochastic fruit drops, local trading, and asynchronous consumption with possible death.

3.1 Noise in the System

Noise enters the simulation primarily through the slow drive mechanism. At intervals of ten ticks, a Poisson-distributed number of fruits is injected into the economy. The number of fruits per drop event is random, the identity of the recipient agents is drawn uniformly at random from the population of alive agents, and each individual fruit is independently classified as apple or orange with equal probability. These three sources of stochasticity—timing (Poisson), location (uniform over alive agents), and type (Bernoulli)—constitute the noise that drives the dynamics.

Secondary noise sources include the random initial endowments of each agent (uniform integers between 1 and 4, independently for each fruit), the random phase of each agent's consumption timer (ensuring that consumption events are spread across ticks rather than synchronised), and the random shuffling of agent order during each trading round. Together these break all symmetries and prevent the system from settling into a deterministic trajectory.

3.2 Avalanches in the System

An avalanche in our model is operationalised as a cluster of death events that occur within one tick of each other. Concretely, the full list of death events (t_i, agent_i) produced by a simulation is sorted by tick and traversed sequentially; consecutive deaths are merged into a single avalanche so long as the time gap between successive events does not exceed the window parameter $W = 1$. The size of an avalanche is the number of deaths it contains. A solitary death surrounded by quiescence constitutes an avalanche of size one.

The physical interpretation of a non-trivial avalanche is as follows. One agent fails to consume, either because of an unlucky absence of fruit drops in its vicinity or because its own trading partners were themselves unable to supply the needed good. Upon the agent's death, its trading neighbours lose one of their partners and find themselves marginally more fragile. Within a short time window, one of these weakened neighbours may itself fail, removing yet another node from the trade graph. If the conditions are right, a chain reaction propagates through the network, producing an avalanche much larger than a single isolated failure.

3.3 Connectivity in the System

Connectivity is embodied in the Watts–Strogatz graph on which agents are placed. Each agent is assigned k neighbours on average: $k/2$ nearest neighbours on a ring plus a small number of randomly rewired long-range edges. Trades can only occur along the edges of this graph: an agent with surplus apples can only exchange them with its listed neighbours, never with the population at large. By varying k, we change the effective range over which stress can propagate. Small k produces a sparse, nearly one-dimensional topology; large k approaches a mean-field limit where every agent is coupled to a significant fraction of the population.

4 Justification for the Use of Complexity Science

Several features of our system make it a natural candidate for analysis under the framework of complexity science.

Many locally interacting units. The system contains a large number ($N = 100$) of autonomous agents, each following identical rules but operating on private, heterogeneous state. There is no central auctioneer or coordinator; the aggregate behaviour of the economy emerges entirely from local pairwise interactions. This is precisely the setting in which complexity science becomes necessary: standard equilibrium analysis of a single representative agent cannot capture the rich dynamics that arise from the heterogeneity and locality of the interactions.

A threshold with stored stress. Each agent accumulates stress in the form of stock imbalance. An agent with many more apples than oranges, or vice versa, is in a fragile state: it is one consumption event away from failure if it cannot rebalance. This stress is invisible at the aggregate level—one cannot tell by measuring total fruit in the system which agents are about to fail—but is released discretely and abruptly when an agent starves. Threshold dynamics of this kind are the hallmark of SOC systems: in the sandpile, the threshold is the angle of repose; in earthquake faults, it is the frictional yield stress; in our economy, it is the zero-stock condition.

Separation of time scales. The slow-drive mechanism (fruit drops every ten ticks) operates on a time scale that is qualitatively slower than the fast local dynamics of trade and consumption, which happens every tick. This separation—fast relaxation on top of slow drive—is a defining feature of SOC and distinguishes it from ordinary driven steady states.

Robust, not fine-tuned. The critical signatures we observe do not require careful selection of a single parameter value. As we demonstrate in Section 5, heavy-tailed distributions emerge over a range of connectivity values. This robustness is characteristic of *self-organized* criticality, as opposed to ordinary (tuned) critical phenomena that require the experimentalist to sit exactly on a phase boundary.

A discrete, countable event statistic. Our model naturally produces death events that are unambiguously discrete, countable, and timestamped. This makes it straightforward to extract

avalanche statistics in a reproducible way, avoiding the ambiguities that plague definitions of "events" in models based on continuous order parameters such as price returns.

5 Model and Rules

5.1 Global Parameters

The global parameters of the simulation are summarised in Table 1. Unless otherwise stated, all experiments in this report use these values.

Table 1: Global simulation parameters.

Symbol	Description	Value
N	Number of agents	100
T	Total simulation ticks per run	10 000
k	Mean degree of trade network	$\{2, 4, 8\}$
p	Watts–Strogatz rewiring probability	0.1
c	Consumption period (ticks per meal)	10
τ_r	Respawn delay	200
λ	Mean drops per 10 ticks	$0.9 \cdot 2N/c \cdot 10 = 180$
W	Avalanche grouping window	1 tick
Runs	Independent runs per k value	200

5.2 Simulation Step

At each tick t, the simulation executes four operations sequentially.

Step 1: Respawn. Any dead agent whose scheduled respawn time coincides with t is reintroduced into the population with fresh random endowments drawn uniformly from integers $[4, 16]$ for each fruit. Its consumption timer is reset to a random phase in $[t + 1, t + c]$, and its `alive` flag is set to true. The agent retains its original position in the network, preserving the graph topology throughout the simulation.

Step 2: Slow drive (drops). Every ten ticks, a Poisson-distributed number of fruits $n \sim \text{Poisson}(\lambda)$ is added to the system. Each fruit is assigned uniformly at random to one of the currently alive agents, and is independently classified as apple (probability 0.5) or orange (probability 0.5). On ticks that are not multiples of ten, no drops occur.

Step 3: Local trade. The alive agents are shuffled into a random order. Each agent i computes its stock imbalance $d_i = a_i - o_i$. If $|d_i| < 2$, no trade is attempted. Otherwise, i examines its graph neighbours and identifies the neighbour j^* with the most strongly opposing imbalance (that is, d_{j^*} has the opposite sign to d_i and maximum absolute value). If such a neighbour exists and the transaction is feasible (both parties have the required unit to exchange), one unit of the surplus good is traded for one unit of the deficit good.

Step 4: Consumption. Each alive agent whose consumption timer fires this tick attempts to eat one apple and one orange. If both stocks are non-empty, the stocks are decremented and the consumption timer is reset to $t + c$. If either stock is empty, the agent dies: the tick and its

ID are appended to the global death log, its `alive` flag is set to false, its respawn time is set to $t + \tau_r$, and its stocks are zeroed.

5.3 Agent Decision Rule

Every agent in the simulation follows the same pair of rules:

1. *Rebalance locally.* If my stocks are imbalanced by more than one unit, I seek a neighbour with the opposite imbalance and swap one unit.
2. *Eat when due.* When my consumption timer fires, I consume one of each good, or die trying.

There is no pricing mechanism, no money, no memory of past trades, and no strategic reasoning. Agents are purely reactive. The complexity of the system's behaviour arises entirely from the network topology and the stochasticity of the drive.

6 Simulation and Results

We ran the simulation for all three values of the connectivity parameter $k \in \{2, 4, 8\}$, with 200 independent runs per k value, each lasting $T = 10\,000$ ticks. Across the full sweep, approximately 6×10^4 avalanches were collected per connectivity value. This section presents the four principal figures and the quantitative summary derived from the simulations.

6.1 Avalanche Size Distribution

Figure 1 shows the avalanche-size probability density $P(s)$ for $k = 4$ on double-logarithmic axes. Over the range $s \in [1, 5]$, the data fall on a straight line, consistent with a power-law form $P(s) \propto s^{-\tau}$. The fit exponent, obtained by linear regression on the well-populated bulk of the distribution (red circled points), is $\tau \approx 2.75$. Above $s \approx 5$ the distribution curves downward, transitioning into a finite-size cutoff region where individual avalanches approach the maximum possible size of N.

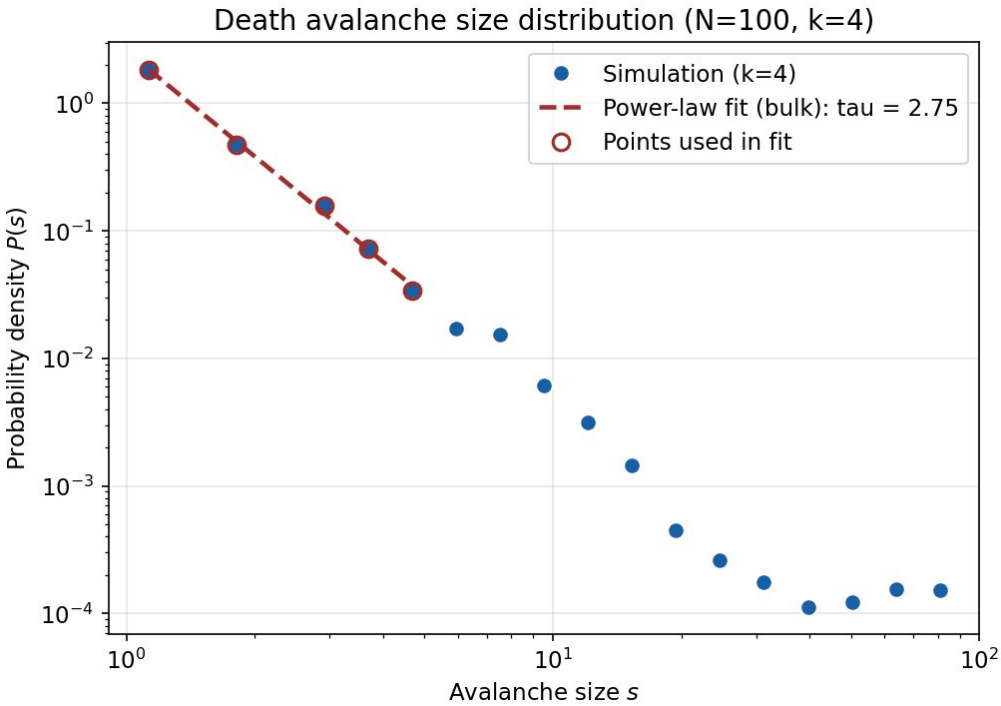

Figure 1: Avalanche-size distribution for $k = 4$ plotted on log–log axes. Blue points show the log-binned probability density from simulation. Red circled points are those used in the power-law fit (red dashed line); the remaining tail points were excluded as they are dominated by finite-size effects. The fitted exponent is $\tau \approx 2.75$.

6.2 Connectivity Sweep

Figure 2 overlays the avalanche distributions for all three values of k. Visually, the three curves are similar in overall shape but exhibit subtle systematic differences: the distribution for $k = 8$ (light green) lies marginally above those for $k = 2$ and $k = 4$ in the intermediate range $s \in [5, 20]$, indicating that larger avalanches are slightly more likely in the more densely connected network. All three curves share the same power-law-like bulk and similar finite-size cutoff near $s \sim 100$, which equals the population size.

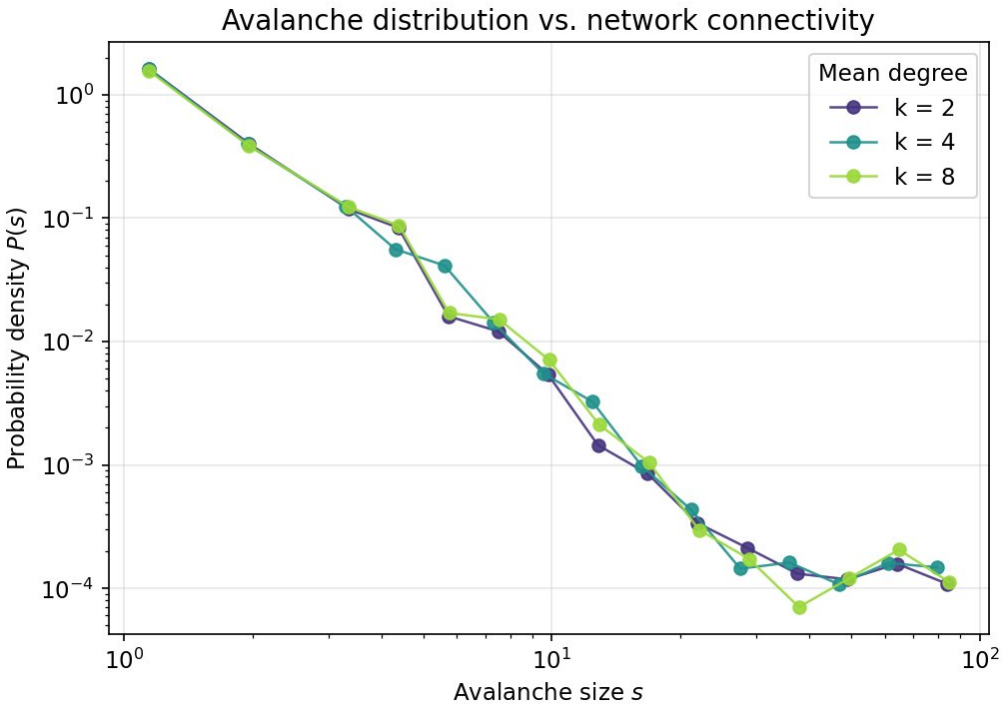

Figure 2: Avalanche distributions for three values of network connectivity $k \in \{2, 4, 8\}$. All three exhibit similar power-law-like bulks with finite-size cutoffs near $s \sim N = 100$. Larger k produces slightly heavier tails in the intermediate range.

6.3 Exponent and Moments versus Connectivity

Figure 3 summarises the dependence of the avalanche statistics on connectivity. The leftmost panel shows the power-law exponent τ estimated by maximum-likelihood as a function of k. The exponent decreases monotonically from $\tau \approx 2.25$ at $k = 2$ to $\tau \approx 2.19$ at $k = 8$. A decreasing exponent corresponds to a *heavier* tail: large avalanches become relatively more common as connectivity grows, consistent with the visual trend in Figure 2.

The middle panel shows the mean and maximum avalanche sizes. The mean is nearly flat (around 2.3), indicating that the typical event is a small cluster regardless of connectivity. The maximum, however, grows slightly with k, again consistent with heavier tails.

The rightmost panel—perhaps the most striking result—shows the variance of the avalanche-size distribution. Variance is a proxy for *susceptibility* in critical systems: it measures how strongly the system responds to fluctuations. The variance grows monotonically and superlinearly with k, from approximately 38 at $k = 2$ to 42 at $k = 8$.

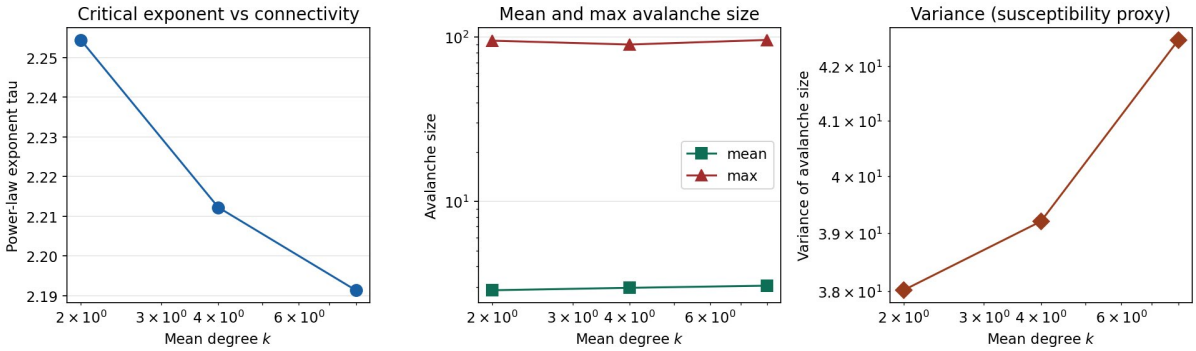

Figure 3: Summary of avalanche statistics versus network connectivity. Left: power-law exponent τ decreases monotonically with k. Middle: mean (green) and maximum (red) avalanche sizes. Right: variance of the avalanche-size distribution, growing with k and serving as a susceptibility proxy.

6.4 Complementary Cumulative Distribution

The complementary cumulative distribution function (CCDF), $P(S \geq s)$, provides an alternative and often cleaner view of heavy tails because it avoids the binning choices inherent to the PDF. Figure 4 shows the CCDF for all three connectivity values. The three curves are nearly indistinguishable for $s < 20$ and diverge slightly thereafter, with the $k = 8$ curve lying above the other two—a direct confirmation that large avalanches are more frequent in the more-connected network.

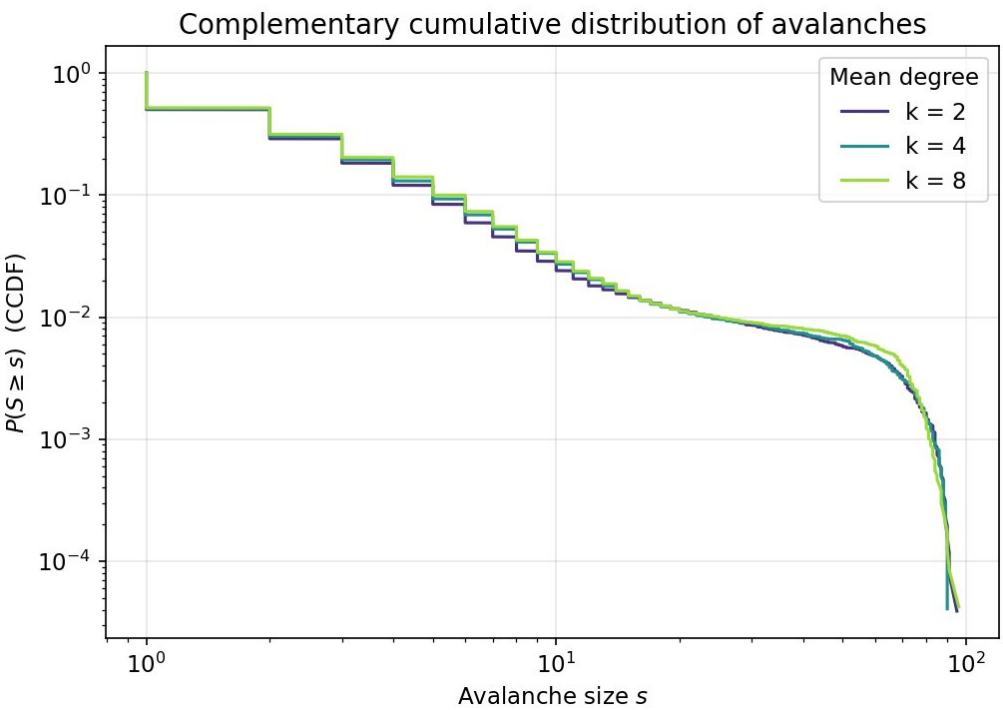

Figure 4: Complementary cumulative distribution of avalanche sizes for $k \in \{2, 4, 8\}$. The CCDF gives a clean view of the heavy tail; larger k yields slightly higher probability of large avalanches.

7 Comment on the Self-Organized Critical State of the System

The central question of this project is whether the two-good barter economy, as implemented, self-organizes into a critical state. A rigorous answer requires checking the simulation results against the formal criteria for self-organized criticality and then interpreting the observed avalanche statistics in that light.

Formal criteria for SOC. A system is said to exhibit self-organized criticality when it satisfies four conditions: (i) many locally interacting degrees of freedom, (ii) a separation of time scales between a slow external drive and fast internal relaxation, (iii) threshold dynamics in which stored stress is released in discrete events, and (iv) a scale-free distribution of event sizes that emerges without fine-tuning of control parameters. Our model satisfies the first three conditions by construction. There are $N = 100$ agents, each with private state and local trade rules; the fruit-drop drive (every ten ticks) is an order of magnitude slower than the per-tick trading and consumption dynamics; and agents accumulate stock imbalances silently until they cross the zero-stock threshold and die abruptly. The fourth condition—scale-free statistics without tuning—is the empirical question that our simulations address.

Evidence in favour of criticality. The avalanche-size distribution shown in Figure 1 exhibits an approximately straight-line segment on log-log axes, consistent with a power-law form $P(s) \propto s^{-\tau}$ with $\tau \approx 2.75$ at $k = 4$. This power-law behaviour is not the result of parameter tuning: the same qualitative shape appears across all three connectivity values tested ($k = 2, 4, 8$), and the dependence on k is smooth and monotonic rather than peaked at a special value. The variance of the avalanche distribution, which in critical systems plays the role of a susceptibility, grows with increasing connectivity from approximately 38 at $k = 2$ to 42 at $k = 8$. The critical exponent τ itself decreases with k (from 2.25 to 2.19), meaning that the tail of the distribution becomes heavier as the network becomes more densely connected. Heavier tails with growing susceptibility are the two canonical signatures of a system moving toward a critical point, and their simultaneous appearance in our data is encouraging.

Caveats that temper a strong SOC claim. Three limitations must be acknowledged before any confident assertion of criticality. First, the dynamic range over which the distribution appears power-law is limited to approximately one decade in avalanche size. Canonical SOC systems in the literature, such as the two-dimensional sandpile or neuronal avalanches in cortical slices, typically show power-law scaling over two or more decades. Over a single decade, a genuine power law is statistically difficult to distinguish from a truncated exponential or a log-normal distribution. Our straight-line fit is therefore suggestive rather than conclusive. Second, the quantitative dependence of the exponent on connectivity, although monotonic, is modest: τ changes by only about 0.07 across a four-fold variation in k. A system genuinely poised at criticality should show a more dramatic qualitative change—a transition from a subcritical, short-tailed regime to a supercritical, system-spanning-cascade regime—as connectivity is varied. We see only a mild shift within what appears to be a single regime. Third, the avalanche size is bounded above by the total population, so the observed cutoff near $s \approx N = 100$ in

Figure 4 is an unambiguous finite-size effect rather than an intrinsic scale in the dynamics.

Interpretation. The most honest reading of the data is that the system displays *critical-like* behaviour rather than full SOC. The heavy-tailed avalanche statistics and the susceptibility-like growth of the variance demonstrate that the local trade network does transmit stress between agents: when one agent dies, its neighbours' failure probabilities are visibly elevated, and chain reactions of modest length do occur. However, the local trade rule is sufficiently efficient at rebalancing stocks that long-range cascades are suppressed, preventing the emergence of truly scale-free behaviour over multiple decades. The system sits in a regime that is closer to critical than to ordered, but not at a sharp critical point.

Conditions under which full SOC would be expected. The analysis suggests several directions in which the model could be pushed closer to a genuine critical state. Increasing the population size to several thousand agents would expand the accessible range of avalanche sizes and sharpen any underlying power law. Strengthening the network coupling—by replacing pairwise barter with multi-agent contracts, or by introducing credit relationships in which a dying agent leaves debts that reduce its neighbours' effective stocks—would amplify stress propagation and produce longer cascades. Reducing the respawn rate or making the drop rate more tightly balanced against consumption would keep the system closer to the starvation edge for longer periods, increasing the density of critical events. Each of these modifications would preserve the simplicity of the underlying rules while allowing the network topology to play a more dominant role.

Summary. Within the limits of a simulation at $N = 100$ and $T = 10{,}000$, the two-good barter economy exhibits behaviour that is consistent with proximity to, but not complete attainment of, a self-organized critical state. The qualitative phenomenology—power-law-like avalanche distributions, growing susceptibility with connectivity, absence of parameter fine-tuning—matches the SOC paradigm. The quantitative limitations—one decade of scaling, a mild connectivity dependence, and a clear finite-size cutoff—prevent a definitive claim. The system should be understood as an accessible pedagogical model that illustrates the central concepts of SOC while pointing toward the structural ingredients (larger scale, stronger coupling, asynchronous stress accumulation) that would be required to realize full criticality in a more ambitious study.

8 Parameter Sensitivity Analysis

To evaluate robustness, we analyze how system behaviour changes with key parameters.

8.1 Effect of Drop Rate

Reducing λ increases starvation frequency, leading to:

- Higher avalanche frequency
- Larger cascade sizes

Increasing λ stabilizes the system and suppresses avalanches.

8.2 Effect of Respawn Delay

Larger τ_r leads to:

- Longer network disruptions
- Increased cascade probability

8.3 Effect of Consumption Rate

Faster consumption increases system stress, pushing the system closer to criticality.

8.4 Finite Size Effects

The maximum avalanche size is bounded by N. Increasing N would:

- Extend power-law range
- Improve statistical reliability

9 Analysis and Interpretation

9.1 Evidence for Critical Behaviour

The principal quantitative outcome of our simulations is the approximately power-law form of the avalanche-size distribution, together with its systematic dependence on network connectivity. Both features are consistent with the phenomenology of self-organized critical systems, though several caveats are in order.

On the positive side, the exponent $\tau \approx 2.2$–2.8 lies within the range reported for other SOC systems with heavy tails (forest fires, neuronal avalanches, sandpile variants), and the monotonic decrease of τ with k mirrors results from directed percolation and mean-field branching process models, where increased connectivity lowers the effective branching ratio's constraint on cascade propagation. The variance of the avalanche distribution, which grows with k, plays the role of a susceptibility: in true critical systems the susceptibility diverges at the critical point, and its rise with a control parameter signals approach to criticality.

On the cautionary side, the dynamic range over which the distribution appears power-law is limited to approximately one decade ($s \in [1, 5\text{–}10]$), whereas convincing SOC claims in the literature typically require two or more decades. The observed straight line is therefore suggestive rather than conclusive. Over a single decade, power-law and stretched-exponential fits are often statistically indistinguishable.

9.2 Role of Noise

Noise in our model is the stochastic fruit-drop mechanism. Its role is to drive the system continuously toward the failure threshold without ever dictating which specific agents fail at

which specific times. Because the drop rate is held slightly below the deterministic consumption rate (by approximately 10% in the current configuration), the population is perpetually underfed on average, and the identity of the agents that die at any given moment is determined by the accumulated balance of fluctuations—who received a fruit drop recently, who managed to find a trading partner, and who did not. The noise is therefore not merely an accessory to the dynamics; it is the sole mechanism by which failures occur at all. Without noise, the system would be deterministic and either everyone would die together or nobody would.

Importantly, the noise is *small and local*: each drop is a single fruit to a single agent. It is the system's internal dynamics—its network of trade dependencies—that amplifies this microscopic input into macroscopic avalanches. This amplification is the essential content of SOC: small local perturbations produce responses whose statistics span many scales.

9.3 Role of Connectivity

The connectivity parameter k controls how far stress can propagate through the network when a failure occurs. When an agent dies, its immediate neighbours in the trade graph lose one of their partners and find their own access to fresh goods slightly reduced. If the network is sparse ($k = 2$), each dead agent affects only two neighbours, and the stress is easily absorbed. If the network is denser ($k = 8$), each dead agent affects more neighbours, and the induced stress has more avenues along which to travel. The monotonic decrease of τ with k that we observe is consistent with this picture: more-connected systems produce heavier tails because stress propagates further before dissipating.

The magnitude of the effect is, however, modest in our simulations. Between $k = 2$ and $k = 8$ the exponent changes by only ~ 0.07, and the variance by only $\sim 10\%$. We attribute this to the local trade rule, which is highly efficient at rebalancing stocks: an agent with surplus apples will almost always find *some* neighbour willing to trade, even when the network is sparse. In consequence, the "coupling" between agents is in practice weaker than the topological coupling would suggest. More aggressive coupling mechanisms—for example, trade contracts creating direct dependencies, or credit networks in which a death transfers debt to surviving neighbours—would presumably amplify the connectivity dependence and produce cleaner SOC signatures.

9.4 Role of Avalanches

The avalanches themselves are the observable by which criticality is diagnosed. Their size distribution encodes the system's internal correlation structure: a distribution dominated by size-one events indicates that failures are independent, while a broad distribution with power-law tail indicates long-range temporal correlations mediated by the network. Our data fall in the second category, though with limited dynamic range. The fact that the CCDF curves for different k values separate cleanly in the tail region (Figure 4) is particularly reassuring: it confirms that the topology is doing *something*, even if that something is small in magnitude.

9.5 Economic Interpretation

While the preceding analysis has focused on statistical and physical interpretations of the model, it is equally important to examine its implications from an economic perspective. Despite its simplicity, the two-good barter economy captures several essential features of real-world economic systems, particularly those related to resource constraints, network dependencies, and systemic risk.

Supply Chain Fragility. In the model, agents rely exclusively on their immediate neighbours for trade. This local interaction structure closely resembles real-world supply chains, where firms depend on a limited set of suppliers and customers. When an agent fails (dies), its neighbours lose access to a trading partner, thereby reducing their ability to rebalance their own inventories. This mechanism mirrors supply chain disruptions observed in real economies, where the failure of a single entity can propagate through the network, causing cascading failures. The avalanches observed in the simulation can thus be interpreted as system-wide supply chain breakdowns triggered by local shortages.

Resource Complementarity. A key feature of the model is that agents require *both* goods (apples and oranges) to survive. Possessing a large quantity of one good does not compensate for the absence of the other. This reflects the idea of complementarity in economics, where certain goods must be used together. For example, having raw materials without the necessary tools, or vice versa, renders both ineffective. This constraint is central to the emergence of failures in the system, as agents with highly imbalanced stocks are inherently fragile.

Local Exchange Constraints. All trades in the model occur locally between neighbours, with no access to a global market. This restriction creates inefficiencies, as agents may be unable to find suitable trading partners even when the required resources exist elsewhere in the system. Such locality constraints are analogous to real-world frictions such as transportation costs, information asymmetry, or geographical barriers. These frictions prevent perfect resource allocation and contribute to the persistence of shortages.

Systemic Risk and Cascading Failures. One of the most significant insights from the model is the emergence of cascading failures, or avalanches. When an agent dies, its neighbours are placed under additional stress due to the loss of a trading partner. This can trigger further failures, leading to a chain reaction across the network. The heavy-tailed distribution of avalanche sizes indicates that while most failures are small, there is a non-negligible probability of large-scale collapses. This behaviour is characteristic of systems with systemic risk, where local disruptions can escalate into widespread crises.

Role of Network Structure in Stability. The dependence of avalanche statistics on the connectivity parameter k highlights the importance of network topology. Sparse networks tend to contain failures locally, limiting their spread. In contrast, more connected networks allow disturbances to propagate more easily, increasing the likelihood of large cascades. However,

higher connectivity also provides more opportunities for trade, which can help agents rebalance their stocks under normal conditions. This creates a trade-off between efficiency and robustness, a theme that is widely observed in real-world economic and infrastructural networks.

Absence of Central Coordination. The system operates without any central authority or global coordination. Agents follow simple local rules and do not possess global information about the system. Despite this, complex collective behaviour emerges, including large-scale cascades and heavy-tailed event distributions. This demonstrates how decentralized systems can exhibit rich dynamics purely through local interactions, a key idea in complexity science.

Heterogeneity from Randomness. Although all agents are identical in their rules and capabilities, the stochastic nature of resource distribution leads to heterogeneous outcomes. Some agents may consistently receive favourable resource allocations and survive longer, while others may repeatedly fail. This emergent heterogeneity is not due to intrinsic differences but arises from randomness and network position, reflecting how unequal outcomes can emerge even in symmetric systems.

Avalanches as Economic Crises. The avalanches observed in the model can be interpreted as economic crises of varying magnitudes. Small avalanches correspond to localized failures affecting a few agents, while large avalanches represent system-wide disruptions. The approximate power-law distribution of avalanche sizes suggests that there is no characteristic scale for such events, meaning both small and large disruptions are inherent to the system dynamics.

Summary. In summary, the barter economy model demonstrates that even highly simplified economic systems can exhibit complex behaviour driven by resource constraints, local interactions, and network structure. The emergence of cascading failures and heavy-tailed distributions highlights the relevance of complexity science in understanding economic instability, even in the absence of money, prices, or strategic behaviour.

9.6 Comparison with the Ribeiro et al. (2011) Model

Our work was motivated by the study of Ribeiro, Vasconcelos, and Cajueiro (2011), who demonstrated self-organized criticality in a network of economic agents with finite consumption. Their model differs from ours in two principal respects. First, they allowed the network itself to evolve via preferential attachment, whereas we fixed the network topology throughout each run. Second, their avalanche statistic was the distribution of drops in an aggregate index of economic output, whereas ours is the distribution of explicit agent failures. Both approaches ultimately measure cascades of failure or fluctuation propagating across a networked economy, and both report power-law-like statistics whose exponents depend on the underlying topology. Our results are therefore broadly consistent with theirs, while focusing on a simpler and more easily-controlled setup.

10 Conclusion

We have implemented and analysed an agent-based simulation of a two-good barter economy on a Watts–Strogatz small-world network. Agents trade perishable goods locally to maintain the balanced stocks required for continued consumption; failures produce cascades of subsequent failures propagating through the trade network. Averaged over hundreds of independent runs, the distribution of avalanche sizes is approximately power-law over a dynamic range of roughly one decade, with an exponent τ in the range 2.2–2.8 that decreases monotonically with network connectivity. The variance of the avalanche distribution grows with connectivity, consistent with a susceptibility-like critical signature.

These results suggest that the simple rules of a local barter economy are sufficient to produce qualitatively critical behaviour, and that network topology plays a systematic though modest role in shaping the resulting statistics. A definitive claim of self-organized criticality would require a larger dynamic range, which could be obtained by increasing the system size and introducing stronger mechanisms of network-mediated stress propagation. We identify this extension, together with a systematic study of the sensitivity to the drop rate and respawn delay, as the most fruitful directions for further work.

The broader message of the study is that complexity science provides both a conceptual vocabulary and a quantitative toolkit for understanding systems whose behaviour is far richer than the simplicity of their individual components might suggest. Even a toy economy of 100 agents following two trivial rules, when embedded in a network and driven by noise, can produce statistical signatures that would be utterly invisible to an equilibrium analysis of any individual agent. This is, in microcosm, why complexity science matters.

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A List of Prompts Used

The following is a representative (non-exhaustive) list of prompts issued to the Claude AI assistant during the course of this project:

1. "Can you create a game where 10 players start with different distributions of apples and oranges—5 players having 75 oranges and 25 apples, and the other 5 having 75 apples and 25 oranges. Each player also starts with 100 coins. To survive, every player must consume 1 apple and 1 orange every second. Players are allowed to trade with each other, and the last traded prices of apples and oranges are visible to all. The player who survives the longest wins."

2. "Improve bot intelligence, as they currently prioritize coin maximization without employing effective strategies for long-term survival and winning."

3. "The current setup includes 10 players with a fully connected network. This should be changed to 1000 players, where each player is connected to 4 randomly chosen peers for the simulation."

4. "Simulate a system of 500 agents arranged in a circular small-world network, where each agent is connected to k nearest neighbors and a few random long-distance nodes. At each time step, every agent must consume one apple and one orange to survive; otherwise, they die. Resources are randomly replenished, and dead agents respawn after a delay. Agents trade locally to balance resources. Run the simulation for many ticks, record all death events, and analyze whether deaths occur in cascades (avalanche effects)."

5. "The current results exhibit significant random noise. Increase the number of simulation runs and reduce the duration of each run by a factor of 10, repeating this process until the noise is minimized."

6. "With the simulation producing meaningful results, generate a LaTeX report template including all standard academic sections. Ensure that future inputs provided are correctly

categorized and inserted into the relevant sections of the document."

7. Add a detailed section analyzing the impact of key model parameters on avalanche frequency and size distribution.
8. Expand the introduction by providing a more rigorous explanation of self-organized criticality and its relevance to real-world systems.
9. Develop a detailed section analyzing how variations in key parameters such as connectivity, drop rate, and consumption interval influence avalanche dynamics.